

Conditional Statements:

- * Simple if $\rightarrow$ Only 1 condition
- * if else $\rightarrow$ Exactly 2 conditions
- * if else if ladder $\rightarrow$ More than 2 conditions
- * Nested if else $\rightarrow$ Nesting of conditions
- * Switch $\rightarrow$ if more than 5 conditions
- * Ternary operator $\rightarrow$ Shorthand if else operator

$n = 0$ ① if ($n > 0$)

② else if ($n < 0$)
positive

③ else
negative
equal to zero

Branching / Nesting

$n = 21$ if (even)
 $\rightarrow$ { if 0 by 4
else ND by 4
else (odd)
ND by 3

$n = 21$ if ($n \% 2 == 0$) {
if ($n \% 4 == 0$) {
n is div by 4
}
else {
n is n div by 4
}
else {
if ($n \% 3 == 0$)
ND by 3
else
not ND by 3
}

Threads (Concurrency)
(5-8 pm) (Scaler)

oops
Exceptions

$\rightarrow$ (JDBC)

Collection Framework

C++

Java

* Standard Template Library
#include <bits/stdc++.h>
map <K>V

* Collections
(java.util)
HashMap

(DSA)

{1, 2, 3, 4, 2, 1}

$\rightarrow$ Bit Masking $O(n)$

0011
0100
0111 = 7

rsb: (rightmost set bit)

= xorAll & -xorAll

= 7 & -7

1010
0010
0001 = 0001

0111
1000
0001
1001

1 3 1
2 4 2
3 1 3
4 2 4
2 1 2
3 1 3
4 2 4
2 1 2

for (int i=0; i < arr.length; i++) {

if (arr[i] & rsb != 0) {

n1 = arr[i] 011131 = 3

} else n2 = arr[i] 0121412 = 4

* Kadane's Algorithm:

Maximum Subarray Sum

arr = {5, -8, 1, 2, -1, 4} i = 6

int cmax = arr[0], 5

int gmax = arr[0], 5

for (int i=1; i < n; i++) {

cmax = max(arr[i], cmax + arr[i]);

gmax = max(cmax, gmax);

} return gmax;

i	arr[i]	cmx+arr[i]	mx a,c	mx cm, gm
1	-8	-8+5	-3	5
2	1	1+(-3)	-2	5
3	2	2+1	3	5
4	-1	-1+3	2	5
5	4	4+2	6	6

* Top level class $\rightarrow$ { private or protected not allowed in Java. }

public or nothing (default) is allowed.

* In C++ $\rightarrow$ by default it is private

* class obj;

* class obj = class();

Object Creation Syntax

8822185799
23092 (Access Modifier Table) (Folder)

Name	Inside Class	Outside Class	Inside Package	Outside Package
public	Yes	Yes	Yes	Yes
private	Yes	No	No	No
protected	Yes	Yes	Yes	Inheritance Yes
default (P ²)	Yes	Yes	Yes	No

Java